

Artista 180 Address Space

The 24-bit address space as this machine actually wires it, read off a full 16 MB memory dump and the on-chip layout in the hardware manual. Only about 4 MB responds; the rest is empty air.

Flash Off-chip SRAM On-chip SRAM On-chip registers Memory-mapped device Observed, purpose unestablished

H'000000 H'002FFF	Boot ROM 12K Exception vectors, reset at <code>boot_reset</code> H'000400, and the serial API — <code>read_serial_data_byte</code>, <code>send_serial_data_byte</code>, <code>get_sci_rx_ready_bit</code>. Byte-identical copies appear at H'008000, H'010000 and H'018000: the device decodes only the low 15 bits, so it repeats every 32K.
H'020000 H'02000F	SED1351F display registers 15 bytes The LCD controller's register window, on its own chip select at the foot of the RAM bank. <code>lcd_controller_init</code> H'20E062 writes H'020001–H'02000F once each, then enables the panel by writing H'DB back to the first. Never read anywhere in the image — write-only. H'020002 = H'4F (79, one less than the 80 bytes per line) and H'020004 = H'EF (239, one less than 240 lines) match the frame buffer exactly.
H'020010 H'03FFFF	External RAM — working storage 128K SCIO receive state and payload buffer live here: <code>sci0_rx_payload</code> H'0203D4, <code>sci0_frame_len</code> H'020886, <code>sci0_rx_state</code> H'02089E.
H'040000 H'044AFF	LCD frame buffer 18.75K 320 × 240 at 2 bits per pixel, MSB first — 19,200 bytes. Behind the Epson SED1351F controller and its two 32K × 8 SRAMs. The upper half of the same bank, from H'048000, holds dense non-display data rather than a second frame buffer.
H'050000 H'05FFFF	Alias of bank 04 64K The same SRAM seen twice: A16 is not decoded, so bank 04 reappears at bank 05, byte for byte across all 64K. Anything written to the frame buffer at H'040000 can be read back at H'050000. A dump therefore captures the screen image twice, which is worth knowing — in the simulator the copy at H'050000 goes stale, because the loader treats the two banks as separate memory and only H'040000 gets redrawn.
H'060000 H'07FFFF	Reads H'FF everywhere 128K Nothing drives the bus here — an idle, pulled-up window.
H'080000 H'09FFFF	Digital input latch 128K window One byte, mirrored at every address in the window because the board ignores the low address lines. <code>input_scan</code> H'209E48 polls it; bits 1 and 2 feed <code>mode_FFEC0</code>. Reads H'70 idle.
H'0A0000	Digital output latch write-only Written by the port-setup code at H'20BF04 onward; reads back as nothing, so it never appears in the dump.
H'0E0000 H'0ECFFF	External RAM — cleared at boot 44K Bulk-written at boot through the copy routine at H'2003A0 — 512 writes per page across the whole span, so this is working memory rather than a device.
H'114000 H'11BFFF	External RAM — application state 32K Where the machine keeps its live state: <code>adc_results</code> H'11A251, the touch debounce counters, the calibration constants copied out of flash at H'11A87E, and the touch position in pixels at <code>touch_x_pixels</code> H'11B102.
H'11F000 H'11FFFF	External RAM — hot page 4K The most heavily written page in the machine: 15.9 million writes in 40 million instructions.
H'200000 H'3FFFFF	Application flash 2M The interrupt handler table sits at H'200000 — the boot ROM's trampolines fetch real handlers from it — followed by the firmware itself, roughly 71,500 instructions reached by static analysis, and the icon bitmaps. Version string <code>MMM03.01</code> at H'200100.
H'500000 H'5FFFFF	Data flash 1M Programmed in place — the firmware issues the AMD unlock sequence to H'575555 and H'572AAA. Holds the configuration block at H'57FF80: the H'A5 validity marker, the four touch-calibration floats at H'57FFA0, and the version string.

THE ON-CHIP PAGE, H'FFF710 – H'FFFFFF

H'FFF710 – H'FFFF0F	On-chip SRAM, 2K. The stack starts at H'FFFD4 and grows down — <code>boot_reset</code> 's first instruction sets it. Machine state lives at the top: <code>CHAN_SELECTION</code> H'FFFD1C, <code>input_flags</code> H'FFFE1, and the touch coordinates <code>touch_x_raw</code> H'FFFD9 / <code>touch_y_raw</code> H'FFFEA that the calibration turns into pixels.	2,048 B
H'FFFF10 – H'FFFF1B	Reserved.	12 B
H'FFFF1C – H'FFFFFF	On-chip registers: DMAC H'FFFF20, ITU H'FFFF60, ports H'FFFC5, SCI0 and SCI1 H'FFFFB0, A/D H'FFFE0, bus controller H'FFFE1.	228 B

CPU REGISTERS

The H8/300H programming model is eight 32-bit general registers, a 24-bit program counter and an 8-bit condition code register. Everything else the CPU controls is memory-mapped into the last 228 bytes of the address space, listed below it.

GENERAL REGISTERS			
Each 32-bit register is addressable in four widths, and they overlap: R0L is the low byte of R0, which is the low half of ER0. ER7 is the stack pointer — this firmware sets it to H'FFFD4 in its first instruction.			
32-BIT	UPPER 16	LOWER 16	HIGH BYTE
ER0	E0	R0	R0H
ER1	E1	R1	R1H
ER2	E2	R2	R2H
ER3	E3	R3	R3H
ER4	E4	R4	R4H
ER5	E5	R5	R5H
ER6	E6	R6	R6H
ER7 stack pointer	E7	R7	R7H

CONTROL REGISTERS	
PC	24-bit program counter. Advanced mode uses all 24 bits; the top byte of a pus discarded.
CCR	Condition codes, bit 7 down to bit 0: I UI H U N Z V C. I masks interrupts arithmetic flags the conditional branches test.

MEMORY-MAPPED PERIPHERAL REGISTERS

All at H'FFFF00 + the offset shown. The SCI, ITU, DMA, I/O port and A/D blocks are the ones this machine actually drives; the simulator models those five.

DMA CONTROLLER H'FFFF20–H'FFFF5F		ITU — 16-BIT TIMERS H'FFFF60–H'FFFF9F		TPC — PATTERN OUTPUT H'FFFFA0–H'FFFFA5		WATCHDOG TIMER H'FFFFA8–H'FFFFA9	
20 MAR0AR	40 MAR2AR	60 TSTR	80 GRB2H	A0 TPMR	A3 NDERA	A8 WDT_TCSR	A9 WDT_TCNT
21 MAR0AE	41 MAR2AE	61 TSNC	81 GRB2L	A1 TPCR	A4 NDRB		
22 MAR0AH	42 MAR2AH	62 TMDR	82 TCR3	A2 NDERB	A5 NDRA		
23 MAR0AL	43 MAR2AL	63 TFCR	83 TIOR3				
24 ETCR0AH	44 ETCR2AH	64 TCR0	84 TIER3				
25 ETCR0AL	45 ETCR2AL	65 TIOR0	85 TSR3				
26 IOAR0A	46 IOAR2A	66 TIER0	86 TCNT3H				
27 DTCR0A	47 DTCR2A	67 TSR0	87 TCNT3L				
28 MAR0BR	48 MAR2BR	68 TCNT0H	88 GRA3H				
29 MAR0BE	49 MAR2BE	69 TCNT0L	89 GRA3L				
2A MAR0BH	4A MAR2BH	6A GRA0H	8A GRB3H				
2B MAR0BL	4B MAR2BL	6B GRA0L	8B GRB3L				
2C ETCR0BH	4C ETCR2BH	6C GRB0H	8C BRA3H				
2D ETCR0BL	4D ETCR2BL	6D GRB0L	8D BRA3L				
2E IOAR0B	4E IOAR2B	6E TCR1	8E BRB3H				
2F DTCR0B	4F DTCR2B	6F TIOR1	8F BRB3L				
30 MAR1AR	50 MAR3AR	70 TIER1	90 T0ER				
31 MAR1AE	51 MAR3AE	71 TSR1	91 T0CR				
32 MAR1AH	52 MAR3AH	72 TCNT1H	92 TCR4				
33 MAR1AL	53 MAR3AL	73 TCNT1L	93 TIOR4				
34 ETCR1AH	54 ETCR3AH	74 GRA1H	94 TIER4				
35 ETCR1AL	55 ETCR3AL	75 GRA1L	95 TSR4				
36 IOAR1A	56 IOAR3A	76 GRB1H	96 TCNT4H				
37 DTCR1A	57 DTCR3A	77 GRB1L	97 TCNT4L				
38 MAR1BR	58 MAR3BR	78 TCR2	98 GRA4H				
39 MAR1BE	59 MAR3BE	79 TIOR2	99 GRA4L				
3A MAR1BH	5A MAR3BH	7A TIER2	9A GRB4H				
3B MAR1BL	5B MAR3BL	7B TSR2	9B GRB4L				
3C ETCR1BH	5C ETCR3BH	7C TCNT2H	9C BRA4H				
3D ETCR1BL	5D ETCR3BL	7D TCNT2L	9D BRA4L				
3E IOAR1B	5E IOAR3B	7E GRA2H	9E BRB4H				
3F DTCR1B	5F DTCR3B	7F GRA2L	9F BRB4L				

REFRESH CONTROLLER H'FFFFAC–H'FFFFAF		SCIO — SERIAL H'FFFFB0–H'FFFFB5		SCI1 — SERIAL H'FFFFB8–H'FFFFBD		I/O PORTS H'FFFFC5–H'FFFFDB	
AC RFSHCR	AE RTCNT	B0 SMR0	B3 TDR0	B8 SMR1	BB TDR1	C5 P4DDR	D1 PADDR
AD RTMCSR	AF RTCOR	B1 BRR0	B4 SSR0	B9 BRR1	BC SSR1	C7 P4DR	D2 P9DR
		B2 SCR0	B5 RDR0	BA SCR1	BD RDR1	C8 P5DDR	D3 PADR
						C9 P6DDR	D4 PBDDR
						CA P5DR	D5 PCDDR
						CB P6DR	D6 PBDR
						CD P8DDR	D7 PCDR
						CE P7DR	DA P4PCR
						CF P8DR	DB P5PCR
						D0 P9DDR	

A/D CONVERTER H'FFFFE0–H'FFFFE9		BUS CONTROLLER H'FFFFEC–H'FFFFEF		SYSTEM AND INTERRUPT CONTROL H'FFFFF1–H'FFFFF9	
E0 ADDRHA	E5 ADDRCL	EC ABWCR	EE WCR	F1 MDCR	F5 IER
E1 ADDRAL	E6 ADDRDH	ED ASTCR	EF WCER	F2 SYSCR	F6 ISR
E2 ADDRBH	E7 ADDRDL			F3 BRCSR	F8 IPRA
E3 ADDRBL	E8 ADCSR			F4 ISCR	F9 IPRB
E4 ADDRCH	E9 ADCR				

SUMMARY			
RANGE	SIZE	KIND	BASIS
H'000000–H'002FFF	12K	Boot ROM	ESTABLISHED
H'020000–H'02000F	15B	SED1351F display registers	ESTABLISHED
H'020010–H'044FFF	148K	Off-chip SRAM, incl. frame buffer	ESTABLISHED
H'050000–H'05FFFF	64K	Alias of bank 04 (A16 undecoded)	ESTABLISHED
H'060000–H'07FFFF	128K	Idle bus	OBSERVED

RANGE	SIZE	KIND	BASIS
H'080000-H'09FFFF	128K	Digital input latch	ESTABLISHED
H'0A0000	—	Digital output latch	ESTABLISHED
H'0E0000-H'0ECFFF	44K	Off-chip SRAM, cleared at boot	ESTABLISHED
H'114000-H'11FFFF	36K	Off-chip SRAM, application state	ESTABLISHED
H'200000-H'3FFFFFF	2M	Application flash	ESTABLISHED
H'500000-H'5FFFFFF	1M	Data flash	ESTABLISHED
H'FFF710-H'FFFF0F	2K	On-chip SRAM	ESTABLISHED
H'FFFF1C-H'FFFFF	228B	On-chip registers	ESTABLISHED

Regions come from scanning Bernina180_20260816.bin at 4K granularity and from watching the firmware run on the simulator; the on-chip boundaries are from the H8/3003 hardware manual. The vertical scale is broken — a linear one would render every mapped region invisible against 16 MB of empty space. Which physical chip covers which range is inferred from where the bus responds, not from the schematic.